

MATH-3108-103-70982-2023FS-Linear Algebra I Exam 3

Michael Colletti

TOTAL POINTS

99 / 102

QUESTION 1

Question 1 6 pts

1.1 Question 1a 3 / 3

✓ - 0 pts Correct

1.2 Question 1b 3 / 3

✓ - 0 pts Correct

QUESTION 2

2 Question 2 3 / 3

✓ - 0 pts Correct

QUESTION 3

3 Question 3 3 / 3

✓ - 0 pts Correct

QUESTION 4

4 Question 4 3 / 3

✓ - 0 pts Correct

QUESTION 5

5 Question 5 3 / 3

✓ - 0 pts Correct

QUESTION 6

6 Question 6 3 / 3

✓ - 0 pts Correct

QUESTION 7

7 Question 7 0 / 3

✓ - 3 pts Incorrect. The correct answer is "Always."

QUESTION 8

8 Question 8 8 / 8

✓ - 0 pts Correct

QUESTION 9

9 Question 9 8 / 8

✓ - 0 pts Correct

QUESTION 10

Question 10 8 pts

10.1 Question 10a 6 / 6

✓ - 0 pts Correct

10.2 Question 10b 2 / 2

✓ - 0 pts Correct

QUESTION 11

11 Question 11 10 / 10

✓ - 0 pts Correct

QUESTION 12

12 Question 12 4 / 4

✓ - 0 pts Correct

QUESTION 13

13 Question 13 16 / 16

✓ - 0 pts Correct

QUESTION 14

14 Question 14 24 / 24

✓ - 0 pts Correct part a)

✓ - 0 pts Correct part b)

✓ - 0 pts Correct part c)

Name: Michael Colletti

Math 3108 – Linear Algebra
Exam 3
Prof. Runnion
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Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has six sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO

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Multiple Choice and True/False. Completely fill in the box corresponding to the correct answer(s). On the multiple choice problems, assume all matrices and vectors contain only real entries.

1. Consider the matrix $A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 5 & 7 & 9 \\ 0 & 0 & 7 & 6 \\ 0 & 0 & 0 & 8 \end{bmatrix}$

(3 points) Is A diagonalizable?

☒ Yes

☐ No

(3 points) How many distinct eigenvalues does A have?

☐ 0

☐ 1

☐ 2

☐ 3

☒ 4

☐ 5

☐ 6

2. (3 points) $\lambda = 0$ is _____ an eigenvalue of an invertible matrix A .

☐ a) always

☐ b) sometimes

☒ c) never

3. (3 points) The vector space $\mathbb{P}_3$ is isomorphic with which of the following vector spaces?

☐ a) $\mathbb{R}^2$

☐ b) $\mathbb{R}^3$

☒ c) $\mathbb{R}^4$

4. (3 points) If a vector space V has a basis with n vectors, then any set in V containing more than n vectors must be linearly dependent.

☒ a) True

☐ b) False

5. (3 points) If a vector space V has dimension n , then any set with fewer than n vectors is automatically linearly independent.

☐ a) True

☒ b) False

6. (3 points) If A is a 4×9 matrix, what is the smallest possible dimension of $\text{Nul } A$?

☐ 0

☐ 1

☐ 2

☐ 3

☐ 4

☒ 5

☐ 6

☐ 7

☐ 8

☐ 9

7. (3 points) If A is an $n \times n$ matrix and $\text{Nul } A = \{\mathbf{0}\}$, then A is _____ invertible.

☐ a) always

☒ b) sometimes

☐ c) never

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Open Answer. Show ALL your work!

8. (8 points) The set $B = \{1+t^2, t-3t^2, 1+t-3t^2\}$ is a basis for $\mathbb{P}_2$. Find the polynomial q

in $\mathbb{P}_2$ given that $[q]_B = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$.

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix} \quad \cdot \quad \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$
$$-1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} -1+0+2 \\ 0+1+2 \\ -1-3-6 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -10 \end{bmatrix}$$

$$q = \boxed{1 + 3t - 10t^2}$$

9. (8 points) The set $B = \left\{ \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$. Find the coordinate vector $[x]_B$ of $x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

$$\left[\begin{array}{cc|c} 4 & 6 & 2 \\ 5 & 7 & 3 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 4 & 6 & 2 \\ 1 & 1 & 1 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 2 & -2 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & -1 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array} \right]$$

$$[x]_B = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

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10. a) (6 points) Find the distance between $\mathbf{u} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$.

$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2}$$

$$\begin{aligned} & \left\| \begin{bmatrix} 10 \\ 6 \\ -8 \end{bmatrix} \right\| = \sqrt{100 + 36 + 64} \\ & = \sqrt{200} \Rightarrow 10\sqrt{2} \end{aligned}$$

distance =

$$10\sqrt{2}$$

- b) (2 points) Are $\mathbf{u}$ and $\mathbf{v}$ orthogonal? ☒ Yes ☐ No (fill in the box)

$\mathbf{u} \cdot \mathbf{v} = 0$
for orthogonal

$$\begin{bmatrix} 12 & 3 & -5 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix} \quad 24 + (-9) + (-15) = 0$$

11. (10 points) Given the matrix $A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & h & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, find the value for h such that the

eigenspace corresponding to $\lambda = 5$ is two-dimensional.

$$\begin{bmatrix} 5-5 & -2 & 6 & -1 \\ 0 & 3-5 & h & 0 \\ 0 & 0 & 5-5 & 4 \\ 0 & 0 & 0 & 1-5 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & -2 & 6 & -1 \\ 0 & -2 & h & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & -4 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & -2 & 6 & -1 \\ 0 & -2 & h & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 & 6 & -1 \\ 0 & 0 & h-6 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & -2 & 6 & 0 \\ 0 & 0 & h-6 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad h=6$$

$$h = 6$$

12. (4 points) Find $\|\mathbf{v}\|$ given $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \\ 0 \\ -2 \end{bmatrix}$.

$$\sqrt{(-1)^2 + (2)^2 + (0)^2 + (-2)^2}$$

$$\sqrt{1+4+4} = \sqrt{9} = 3$$

$$\|\mathbf{v}\| = 3$$

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13. (16 points) Find the complex eigenvalues of $A = \begin{bmatrix} 7 & -5 \\ 1 & 3 \end{bmatrix}$ and find a basis for each eigenspace.

$$\begin{bmatrix} 7-\lambda & -5 \\ 1 & 3-\lambda \end{bmatrix}$$

$$(7-\lambda)(3-\lambda) + 5$$

$$21 + \lambda^2 - 10\lambda + 5 \Rightarrow \lambda^2 - 10\lambda + 26$$

$$\lambda = \frac{10 \pm \sqrt{100 - 4(26)}}{2} \Rightarrow \frac{10 \pm 2i}{2} \Rightarrow \boxed{5 \pm i} \text{ eigenvalues}$$

$$\lambda = 5 + i$$

$$\begin{bmatrix} 7-5-i & -5 & | & 0 \\ 1 & 3-5-i & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2-i & -5 & | & 0 \\ 1 & -2-i & | & 0 \end{bmatrix} \xrightarrow{R_2(2-i)+R_1 \Rightarrow R_2} \begin{bmatrix} 2-i & -5 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$(2-i)x_1 = 5x_2$$

$$\vec{x} = \begin{bmatrix} \frac{2}{5} - \frac{i}{5} \\ 1 \end{bmatrix}$$

& complex conjugate

$$\vec{x} = \begin{bmatrix} \frac{2}{5} + \frac{i}{5} \\ 1 \end{bmatrix}$$

$$\lambda = 5 - i$$

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14. Consider the matrix $A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{bmatrix}$ which has eigenvalues $\lambda = 1$ and $\lambda = 5$.

a) (8 points) Find a basis for the eigenspace of A corresponding to $\lambda = 1$.

$$\begin{bmatrix} 2-1 & 2 & -1 \\ 1 & 3-1 & -1 \\ -1 & -2 & 2-1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & -1 \\ -1 & -2 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$x_1 = -2x_2 + x_3$$

$$x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{basis} \Rightarrow \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

b) (8 points) Find a basis for the eigenspace of A corresponding to $\lambda = 5$.

$$\begin{bmatrix} 2-5 & 2 & -1 \\ 1 & 3-5 & -1 \\ -1 & -2 & 2-5 \end{bmatrix} \Rightarrow \begin{bmatrix} -3 & 2 & -1 \\ 1 & -2 & -1 \\ -1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -1 \\ -1 & -2 & -3 \\ -3 & 2 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & -4 & -4 \\ -3 & 2 & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & -4 & -4 \\ 0 & -4 & -4 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 & | & 0 \\ 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$x_1 = -x_3$$

$$x_2 = -x_3$$

$$x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \text{basis}$$

c) (6 points) Diagonalize A . That is, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

$$P = \begin{bmatrix} -2 & 1 & -1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$